VAIBHAV MENARIYA

B.Tech in Electronics & Communication Engineering (ECE) — Semiconductor & VLSI Specialization

JECRC University, Jaipur | In Industry Partnership with TrueChip Solutions

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1. ARCHITECTURAL PROFILE & OBJECTIVE

Motivated first-year Electronics & Communication Engineering undergraduate at JECRC University, pursuing a specialized industry-partnered curriculum in Semiconductor & VLSI System Design with TrueChip Solutions. Solid foundations in digital logic gate modeling, PN junction physics, and procedural programming. Actively developing Python automated hardware telemetry tools and simulation prototypes to bridge the gap between physical silicon design and modern software orchestration.

2. FORMAL EDUCATION & PIPELINE

Bachelor of Technology (B.Tech) — Electronics & Communication Engineering **2026 — 2030 (Current // 1st Year)**

JECRC University, Jaipur — Semiconductor & VLSI Specialization (in partnership with TrueChip)

- Industry-aligned curriculum designed in collaboration with TrueChip Solutions (EDA workflows & verification paradigms).
- Core Coursework: Semiconductor Physics, Digital System Design, Boolean Logic Synthesis, Circuit Analysis, and C Programming.
- Active hands-on lab experimentation with circuit instrumentation, breadboard prototyping, and logic ICs.
- Active participant in university technical clubs, hardware hackathons, and electronics design groups.

Senior Secondary Education (Class XII — RBSE Board) **Completed**

Mahaveer Academy, Udaipur, Rajasthan — Science Stream (Physics, Chemistry, Mathematics)

- Built strong fundamentals in differential & integral calculus, electromagnetic wave theory, semiconductor diodes, and vector algebra.
- Developed disciplined analytical reasoning and mathematical logic foundational to engineering computation.

3. TECHNICAL ARSENAL & SKILLS MATRIX

Hardware & VLSI Core:	Semiconductor Physics, Digital Logic Gates, Boolean Minimization, CMOS Basics, ASIC Design Flow Primer
Programming & Scripts:	Python (Hardware Telemetry, Automation, Async IO), C (Procedural, Bitwise Operations), Shell/Bash
EDA & Simulation Tools:	Multisim, Circuit Simulation, SPICE primers, Logic Analyzer Workbench
Developer Toolchain:	Visual Studio Code, Git, GitHub, Linux / Terminal Environments, Markdown Documentation
Core Engineering:	Algorithmic Logic, Problem Solving, Hardware-Software Integration, Schematic Interpretation

4. KEY ENGINEERING PROJECTS & SYSTEMS

Digital Circuit Simulator & Logic Analyzer (Web-based Engine) Next.js | React | CMOS Logic Models

- Engineered an interactive web-based logic gate simulator modeling standard CMOS gates (AND, OR, XOR, NAND, NOR, XNOR, NOT).
- Features real-time dual-input state manipulation, dynamic truth table row highlighting, and visual voltage state feedback.

Automated Task & Hardware Signal Logger Python 3.x | Asyncio | Telemetry Parsing

- Developed automated Python scripts for capturing, timestamping, and parsing continuous data signal benchmarks.
- Implemented anomaly detection algorithms to identify voltage swing deviations and log telemetry records to structured formats.

Semiconductor Verification Case Study & ASIC Flow Primer SystemVerilog | UVM Concepts | ASIC Lifecycle

- Conducted technical research into modern ASIC flows: RTL Synthesis, Static Timing Analysis (STA), and DRC/LVS physical sign-off.
- Authored primer notes detailing TrueChip's verification methodology and SystemVerilog/UVM constrained-random testing concepts.

5. TECHNICAL INTERESTS & ACTIVITIES

- Low-Latency System Mechanics: Deep exploration into frame pacing, 1% low frame optimization, and high-frequency input polling.
- Robotics & Microcontrollers: Hands-on prototyping with ESP32 and Arduino boards, exploring PWM motor control and I2C/UART sensors.
- Semiconductor Community: Active participant in student technical chapters, electronics workshops, and industry speaker sessions.